



*EMIA6500R (CWB) · CMAA5022 (GZ) ·
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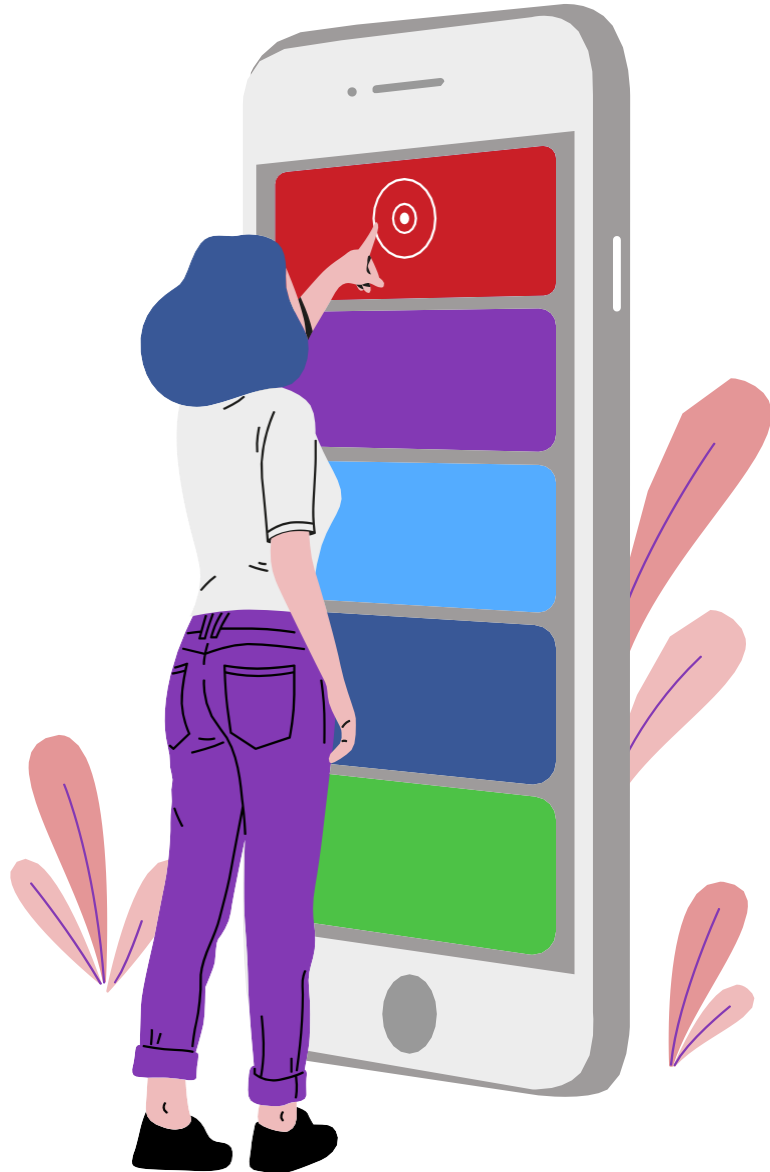
Social Media for Creatives

Lecture 09

Extended Reality and NFT

Professor Pan HUI; FRI 15:00 – 18:00

Today's Outline



01

Lecturing

Extended Reality and NFT

02

Digital Teacher

VR in Social Media

03

Group Presentation

Group 5

04

Group Discussion

VR Miro Time!

05

Seminar Series

Professor Lik-Hang (Paul) Lee, PolyU

Immersive Virtual Environment

- An **immersive virtual environment** refers to an artificial, interactive, computer-created scene or "world" which can **blend elements of the physical and digital worlds** ([Nechvatal, 1999](#)).
- An immersive digital environment could be a model of reality, but it could also be a complete fantasy user interface or abstraction.
- The term “*immersive*” refers to users’ perception of **being physically present in a non-physical world**, which is created by virtual elements like images, sound or other stimuli to provide an engrossing environment.
- Some factors are crucial to create such a perception, including –
 - **Element-based**: Believable 3D computer graphics, surround sound, interactive user-input ([Naef, 2002](#)).
 - **Experience-based**: Similarity between subjects and representation in visual appearance, interactivity to virtual elements ([Slater, 1993](#)).

Extended Reality (XR)



- **Extended Reality (XR)** is an umbrella term that encompasses a range of **technologies** that enable users to experience immersive virtual environments.
- XR is the access to virtual worlds, by allowing users to **interact with or manipulate** objects and **communicate** with other users in virtual environments.
- The fields of XR are **rapidly growing** and being **applied in a wide range of areas** such as entertainment, marketing, real estate, training, and remote work ([Chuah, 2018](#)).
 - For instance, National Football League (NFL) has been using XR to ***train quarterbacks***. The XR training allows quarterbacks to practice their plays and decision-making in a realistic and safe environment ([Source](#)).
- Specifically, XR technologies includes augmented reality (AR), virtual reality (VR), and mixed reality (MR).

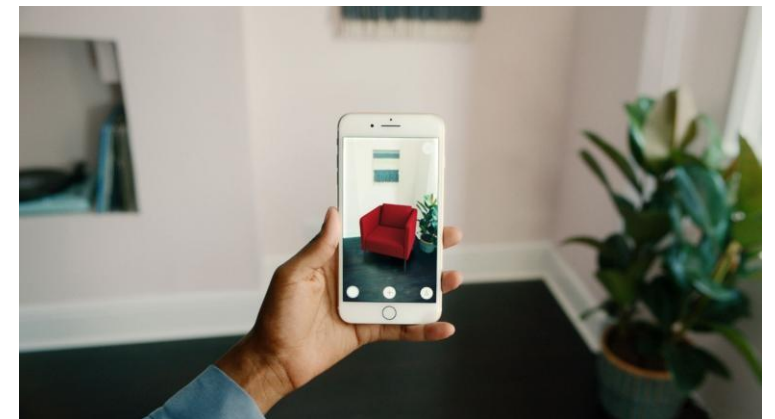
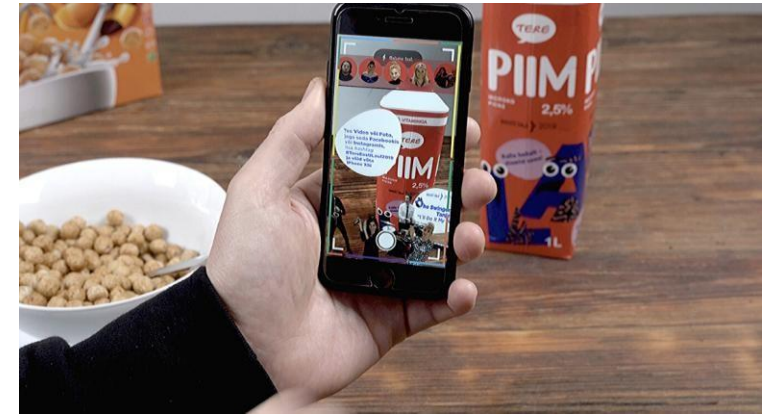
Augmented reality (AR)

- First, we begin with augmented reality.
- **Augmented reality (AR)** is an interactive experience that **combines the real world and computer-generated content**.
 - The content can span multiple sensory modalities, including visual, auditory, haptic, somatosensory and olfactory ([Cipresso, 2018](#)).
- According to relevant literature, AR can be also defined as a **system** that incorporates **three basic features** ([Wu, 2013](#)) –
 - A combination of digital and physical worlds
 - Interactions made in real time
 - Accurate 3D identification of virtual and real objects.
- One noteworthy property of AR is that users' experience is ***seamlessly interwoven*** with the physical world such that it is perceived as an immersive aspect of the real environment ([Azuma, 1997](#)).
 - The physical world is “augmented” by virtual elements but **not completely replaced**.



Type of Augmented Reality (1)

- Researchers have summarized **six** different types AR that fall **under two overarching categories**. These include triggered versus view-based augmentation ([Edwards-Stewart, 2016](#)).
- In **triggered AR**, triggers are stimuli or characteristics that **initiate** or “**trigger**” the augmentation.
 - Triggers can be paper or object markers, GPS location, dynamic augmentations of objects.
 - I.e., QR-Codes as AR markers to generate AR effects.
- Other forms of AR are **view-based**, which includes either digitized augmentations **without reference** to what is in view, or **augmentation of a stored/static view**.
 - For example, the IKEA Place app to recommend matching furniture items to those it recognizes in a customer’s room.



Type of Augmented Reality (2)

- Among triggered AR technologies, four types exist.
- **Marker-based AR:** Marker-based AR requires a marker to activate an augmentation. Such markers can be paper-based or physical objects that exist in the real world.
 - Augmentations related to the marker enhance the image or object, though some are only a means to access digital content.
 - For example, the app Aurasma's augments the appearance of a real-world \$20 bill, which morphs into an entertaining, patriotic animation.
- **Location-based AR:** Location-based AR uses the device's GPS location as a trigger to pair dynamic location.
 - It can also be embedded with relevant data or information matching viewer's interest (e.g., restaurants in Yelp's monocle view).



Type of Augmented Reality (3)

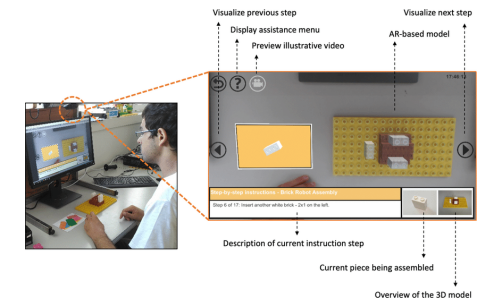
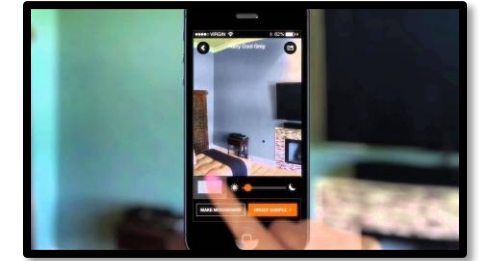
- **Dynamic augmentation:** dynamic augmentation is a type of AR effects responsive to the view of dynamic/changing object.
 - One typical dynamic augmentation is the AR with motion tracking to fit the identified object.
 - For example, Swivel is a shopping application that allows users to try on clothing and accessories digitally.

- **Complex augmentation:** This type of AR is a *combination* of aforementioned three types of AR, with functional assistance via the Internet.
 - This can be seen in the original concept for Google Glass, wherein users see information about local sites based on their GPS location.
 - Objects in the user's field of vision are presented with helpful information about their surroundings from an internet backend.



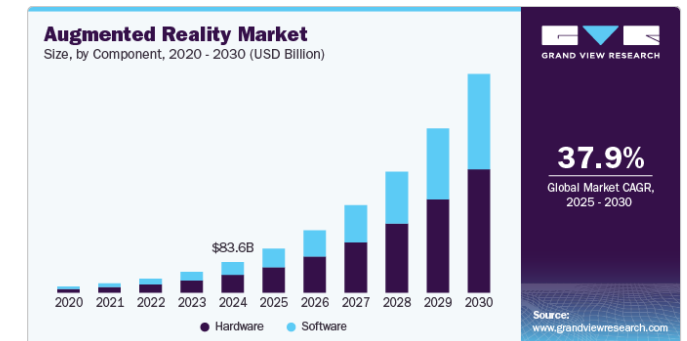
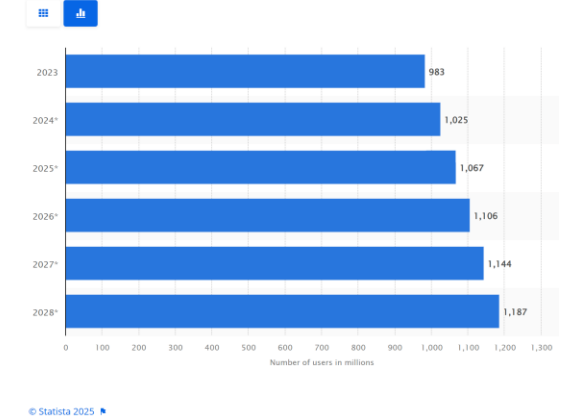
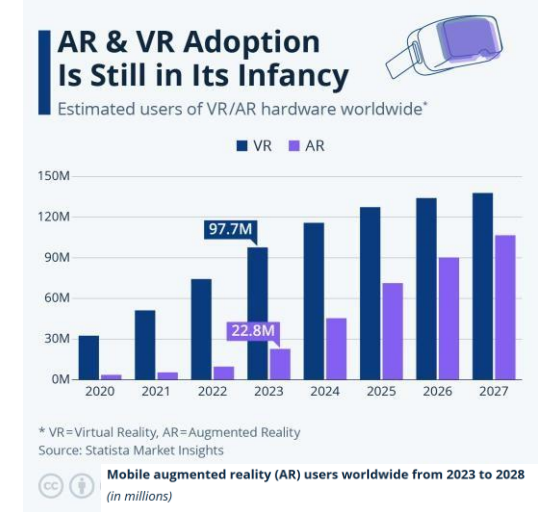
Type of Augmented Reality (4)

- View-based AR includes indirect augmentation and non-specific digital augmentation.
- **Indirect augmentation:** This type of AR presents effects based on statistic or recorded scenes, where the augmented targets are not real-time (e.g., images, videos).
 - An example of this category is Wall Painter, an application allow users to take a picture of a room and then change the wall color. The application can identify the wall from other objects and augments only the wall.
- **Non-specific digital augmentation:** This type of AR can present effects without any physical reference, where the virtual elements may just overlap the view of physical world.
 - This is common in mobile games. The user interacts with the augmentation, such as tapping the augmentation when it comes into view but without reference to the user's environment.



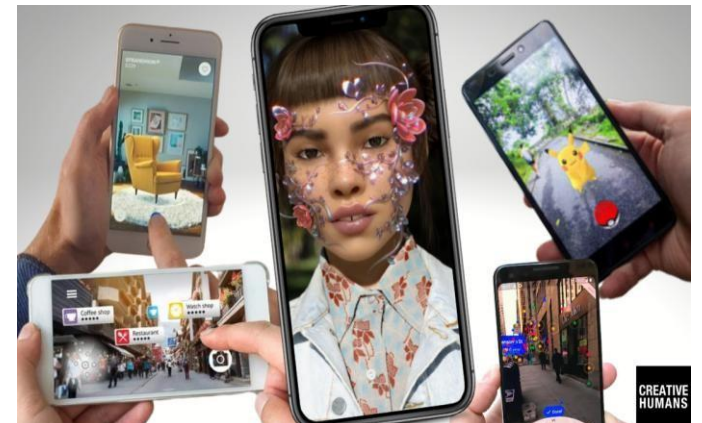
AR on Social Media (1)

- On **social media**, the **AR marketing** landscape is evolving at a rapid pace. The emergence of hardware and software innovations that give AR the potential to disrupt the market and become a mass-market technology ([Rauschnabel, 2021](#)).
 - According to Statista, in 2023, there were 22.8 million AR users globally, and it is estimated that over 1,187 millions Mobile AR users worldwide in 2028 ([Statista](#)).
- Moreover, brands and social media are increasingly **investing** in AR-based marketing tools.
 - It is reported that investment on AR tools development was 83.6 billion in 2024 and is expected to expand at a compound annual growth rate of **37.9%** from 2020 to 2030 ([Source](#)).



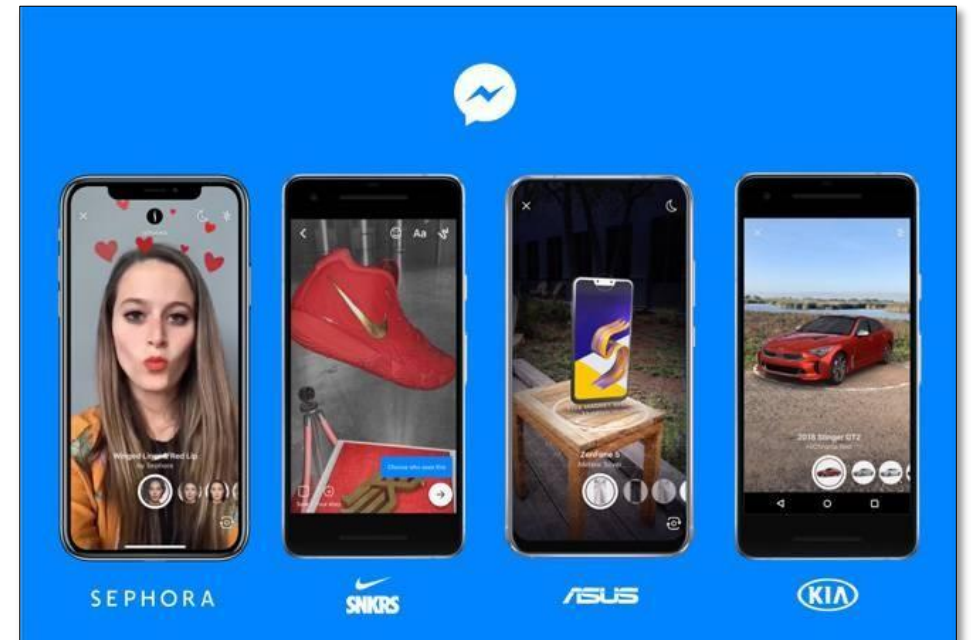
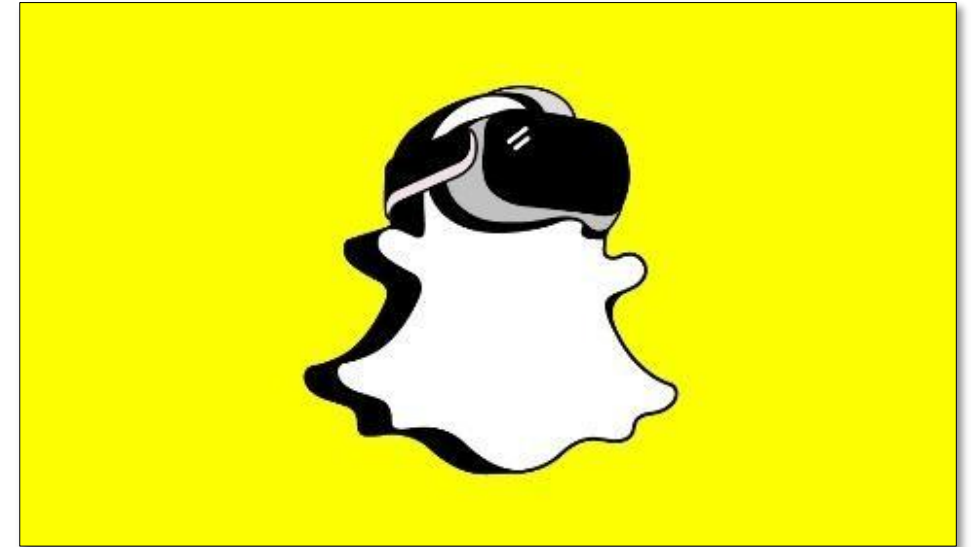
AR on Social Media (2)

- AR applications also have an **impact to online society**. One 2021 report of AR consumers by **snap.com** has revealed several insights ([Source](#)).
- AR is **growing virally** on online society, accompanied with the mobile usage boom. Such a rapid growth is mostly driven by **younger, and multi-person households**.
 - Gen Z are both **71%** more likely to use AR all the time vs. older generations. And multi-person households are **89%** more likely than single-person households to use AR in daily life.
- AR **enhances connection** and interaction between brands and consumers. A lot of people are using AR to make purchase decisions and plan to keep doing so.
 - **On snap.com, 100+** million users have shopped with AR online and **94%** of them agree they'll use AR the same or more when shopping next year.



AR on Social Media (3)

- Social activities in online communities are driving **AR adoption**. People are willing to share their favorite Brand's AR experiences.
 - For Gen Z, **33%** rely on friends and family to discover branded AR experiences and **54%** consider sharing branded AR is an important interaction with friends.
- AR interactions between users also **drive communication**, which facilitates its **spread** on social networks.
 - Around **82%** users report that they learn about AR information from communication networks.
 - Frequent AR users are more likely to have **over 50%** of their friends and family using AR.



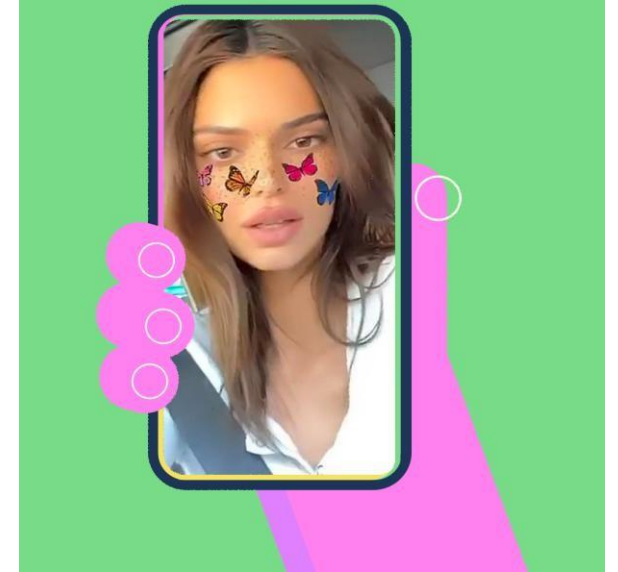
AR Filters on Social Media

- On social media, one popular AR application is the **AR filters**.
- A filter is an **appearance-altering** digital image effects often used on social media.
- Filters initially simulate the effects of camera filters, and nowadays have developed with facial recognition technology and computer-generated AR.
 - On many social media platforms (e.g., Instagram, TikTok, and Snapchat) filters—especially **beauty filters**—are often used to alter the appearance of selfies taken on mobile devices.
- AR filters are digitally based responsive interactions applied to the **user's face or surroundings** to extend or change what is being viewed in the actual environment ([Rios, 2018](#)).
 - A recent report shows that **more than 4.5 billion** AR photos and videos are taken daily by Snapchat users, which stresses the potential of AR filters as a communications and engagement tool ([Source](#)).



Why Choose AR Filters? (1)

- We have discussed the popularity of AR filters on social media. But **why** users are willing to use AR filters?
- To answer this question, Javornik et al. first investigated users' motivation for using AR face filters on **Instagram** ([Javornik, 2022](#)). According to an interview (N=14) and online survey (N=536), they summarize several **motivations as significant drivers of usage behaviors**.
- **Ideal self-presentation**: The self-presentation of an ideal self through AR filters (e.g., selfies or facial transformation with AR effects) significantly drove the use frequency of the filters.
- **Curation**: Users' creative content curation via AR filters was also shown to increase users' positive mood, encourage them to continue using more AR filters.



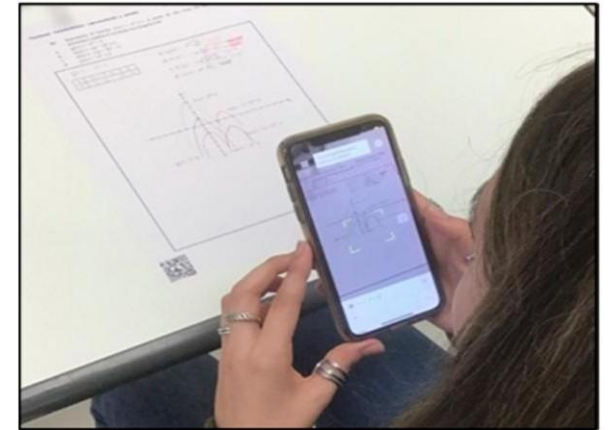
Why Choose AR Filters? (2)

- **The need to social interaction:** The need to engage in social interactions was shown to drive both the filters' exploration as well as use frequency.
- **Enjoyment:** Participants also reported enjoyment as a significant motivation for both exploratory use and use frequency of AR filters. In addition, enjoyment also **reinforces their connections** to the content created by AR filters.
- Moreover, the authors also discussed some relevant insights of AR filters in relation to **well-being** on social relationships.
 - Pursuing true and transformed self-presentation through AR face filters was shown to significantly increase both individual **self-acceptance** and **positive affect**.
 - Creative content curation, interaction, and enjoyment were all shown to **construct a positive mood** for users.



AR Filters Innovation in Education (1)

- AR filters can also be utilized in *child learning*, as an effective tool to **boost AI-based education**.
- Nowadays, many children growing up have access to AI-enabled devices from an early age. However, studies have shown that children **lack understanding** of how their smart toys and AI-assistants work ([Druga, 2017](#)).
- To bridge the gap between children's perception and AI learning tools, Jordan et al. first took a look on the **body-movement-focused interactive AI** techniques such as AR filters ([Jordan, 2021](#)).
- With the motivation to enhance education experience, they proposed a toolkits, called **PoseBlocks**, for children to learn AI knowledges and build interactive projects by body-sensing AR filters.
 - The AR filters can generate animation effects to instruct AI knowledges and guide children to build their projects step-by-step.



AR Filters Innovation in Education (2)

- Their innovation of AR Filters in child education was *successful*.
- According to a user test on young students (aged 9-14, N=34), after only a week-long pilot development, **21(61.7%)** students managed to complete their AI projects with PoseBlocks. This suggests AR filters' effectiveness as an interactive tool to **facilitate learning of AI knowledges**.
 - Students' projects were creative and covered a wide range of fields (e.g., health & well-being, games, education, etc.).
- In addition, the authors also noted **three crucial principles** to apply AR filters in children education –
 1. **Tinkerability**: Developers should consider students' abilities to “experiment with new ideas iteratively”, as well as enabling instantaneous feedback.
 2. **Project-based learning**: Designing and building practical projects helped students digest knowledge effectively.
 3. **Alignment with passions**: Students were more engaged when working on projects that were meaningful to them or aligned with their interests.

Challenges for AR Education (1)

- However, except the success of AR filters, relevant survey paper has listed some major **challenges** for the practical use of AR techniques in educational applications ([Akçayır, 2017](#)).
- **Difficulty of usability**: The most reported challenge is that AR is “difficult for students to use.” **Usability** is an important technical factor, which affects educational effectiveness ([Chang, 2014](#)).
 - For example, without well-designed interfaces, students may experience difficulties when using this technology ([Munoz-Cristobal, 2015](#)).
- **Time-consuming**: Another difficulty related to usability is that AR may cause time loss for students and require excessive additional lecture time.
 - A study reported that an AR-using group in their study required significantly longer mean training times compared to their non-AR-using group ([Gavish, 2015](#)).

Challenges for AR Education (2)

- **Cognitive overload**: It is suggested that students might experience cognitive overload in an AR learning environment due to the amount of material and complexity of tasks, which could increase their pressure during learning ([Cheng, 2013](#)).
- **GPS errors**: This is one of the most experienced problems in location-based AR applications. Global Positioning System (GPS) error is a problem caused by the AR application miss-perceiving a location and/or direction.
 - The low precision of geo-locating and frequent loss of location information can **hinder** the process of learning and **cause frustration** among students ([Dunleavy, 2009](#)).
- **Other challenges**: There are still some noteworthy challenges for AR educators, including low sensitivity in triggering recognition, inability for large-group teaching, distraction of students' attention, and inadequate teacher ability.

Success of AR Game – Pokémon Go (1)

- In addition to education, AR technology has also been widely applied in **mobile game industry**. **Pokémon Go** is one well-known example.
- Pokémon Go is a mobile game that uses AR to allow players to capture, battle, and train virtual creatures known as Pokémon.
- It was launched in 2016 and quickly became a global sensation.
 - According to Statista, released in July 2016, Pokémon GO had accumulated **600+ million** lifetime downloads. During its first quarter, the app generated a dizzying **228 million** downloads ([Statista](#)).
- Pokémon Go is also a great success in AR marketing.
 - In the first month after released, Pokémon Go revenue broke records, at **\$207 million**. In 2022, Pokémon GO in-app purchase revenues surpassed **645.59 million** U.S. dollars worldwide([Statista](#)).



Success of AR Game – Pokémon Go (2)

- ❑ One key reason for the success of Pokémon Go is the way it **enhances users' immersive experience**.
 - Players had to physically move around in their environment to find and capture Pokémon, which created a sense of immersion and excitement by AR technology ([Pyae, 2017](#)).
- ❑ The game also transforms real-world landmarks into virtual interactive facilities (e.g., Pokéstops and gyms) for multi-player activities, encouraging players to **form local communities** and **strengthen connections** with other players ([Dorward, 2016](#)).
- ❑ The social aspect of Pokémon Go also played a significant role in its success. Players could **team up with friends or join online groups** to compete against other teams and participate in special events.
- ❑ This creates a **sense of community** among players and helps to **sustain interest** in the game over time ([Bhattacharya, 2019](#)).



Pokémon Go as Social Phenomenon

- Pokémon Go also impacts both online and physical society.
- The game is referred to as a "**social media phenomenon**" which has brought people together from diverse walks of life, with a sudden emergence of public fever for Pokémon hunting.
 - It is reported that 231 million people engaged in 1.1 billion interactions that sharing hunting experience on Facebook and Instagram in the first month ([Source](#)).
 - Numerous media outlets referred to the surge in popularity as "Pokémon Go Mania", or simply "Pokémonia" ([Source](#)).
- Such a fever later turned into physical world.
- During the first week after the game's release, multiple events and gatherings organized by the players happened across the world.
 - I.e., it was reported that about 5,000 people gathered at Millennium Park, Chicago for hunting Pokémon since the game was released in a couple of weeks ([Source](#)).



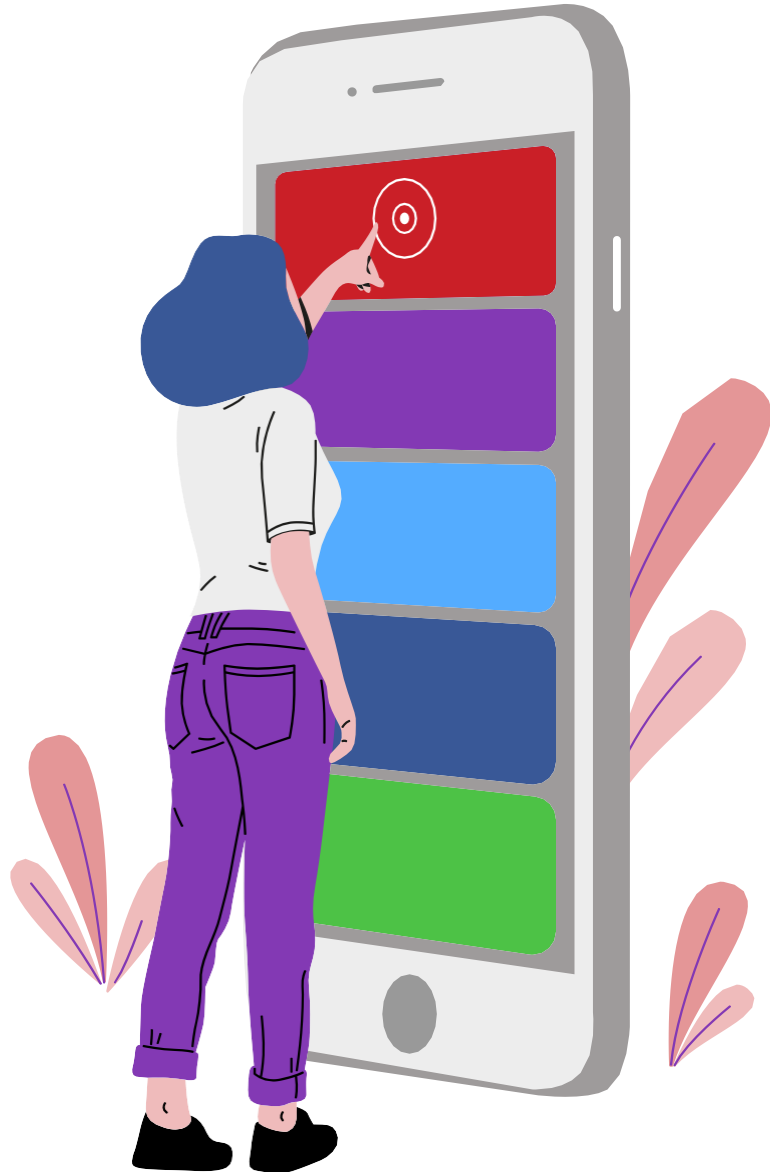
The Risk of Distraction

- However, while Pokémon Go uses AR to create immersive game-play experience for players, it also bring **risk of public accidents by distraction**.
- One actual measurable data by JAMA Internal Medicine suggested that reckless Pokémon hunting really did **distract** drivers and pedestrians.
- The research analyzed tweets and news items to piece together a picture of Pokémon-related traffic incidents ([Source](#)).
 - By the authors' calculations, an estimated **113,993** tweets were sent **within the span of 10 days** that depicted drivers or pedestrians distracted by the game during traffic.
 - The results also showed that, within a 10-day span, there were **14 crashes** reported to involve Pokémon Go.

Right fig records one driver crashed into a tree while playing Pokémon Go on his smartphone in New York ([Source](#)).



Today's Outline



02

Digital Teacher

VR in Social Media

DIANA

- Gender: Female
- Age: 29
- Nationality: German
- One of the two most popular digital teachers in 2024
- Expert in media studies, and the impact of social media on public discourse



Virtual Reality (VR) (1)

- Next, we introduce virtual reality.
- **Virtual reality (VR)** is a simulated experience that employs pose tracking and 3D near-eye displays to give users an immersive feeling of a virtual world.
- VR is featured by its **fully** immersive experience that tricks the user's senses into believing that they are in a different environment.
- Usually, VR technology creates a 3D environment that the user can interact with through a headset and hand-held controllers.
- According to Burdea et al., there are **three** key features to construct VR experience – **interactivity to virtual elements**, **a sense of immersion**, and **imagination from human's views** ([Burdea, 2003](#))



Virtual Reality (VR) (2)

- The virtual reality (VR) market has been **growing rapidly** in recent years, driven by advancements in technology, increased investment, and a growing demand for immersive experiences in various industries.
 - It has been reported that the industry is growing at a fast pace, with the global VR market size projected to increase from less than **12 billion** U.S. dollars in 2022 to more than **\$22 billion** by 2025 ([Statista](#)).
- The market is expected to be driven by the increasing demand for VR in **gaming and entertainment**, which is the largest contributor to the VR market.
 - It is reported that VR gaming industry accounts for **more than 50%** of the market share.
 - Moreover, global revenue within the VR gaming industry is forecast to grow to **\$2.5 billion** in 2023, up from **\$1.4 billion** in 2021 ([Statista](#)).



Virtual Reality (VR) (3)

- As an emerging immersive technology, VR has also shown its **effectiveness** and **gained popularity** in many science fields.
- In social sciences and psychology, VR offers a **cost-effective** tool to study and replicate interactions in a controlled environment. It can be used as a form of **therapeutic intervention**.
 - For instance, there is the case of the **VR exposure therapy (VRET)**, a form of exposure therapy for **treating anxiety disorders** such as post traumatic stress disorder (PTSD) and phobias ([Gonçalves, 2012](#)).
- In **medicine**, one crucial VR application is simulated VR **surgical environments**. Under the supervision of experts, VR can provide **repeatable training** at a low cost, allowing trainees to recognize and amend errors as they occur ([Haque, 2002](#)).



Social VR

- **Social VR** is a type of virtual reality experience that allows multiple users to interact with each other in a shared virtual space.
- Compared to traditional VR platforms, social VR platforms is characterized by its ability to **enable social communication and interaction**, where forming social relationship and communities is possible ([Lee, 2021](#)).
- According to Liu and Steed, the usability of a social VR are significantly affected by **three** main components ([Liu, 2021](#)), including –
 - **Interactive interface** for users to conduct social activities.
 - Functions to support **avatar creation, manipulation, and self-expression**.
 - Mechanisms to **manage social relationship or networks** between users.



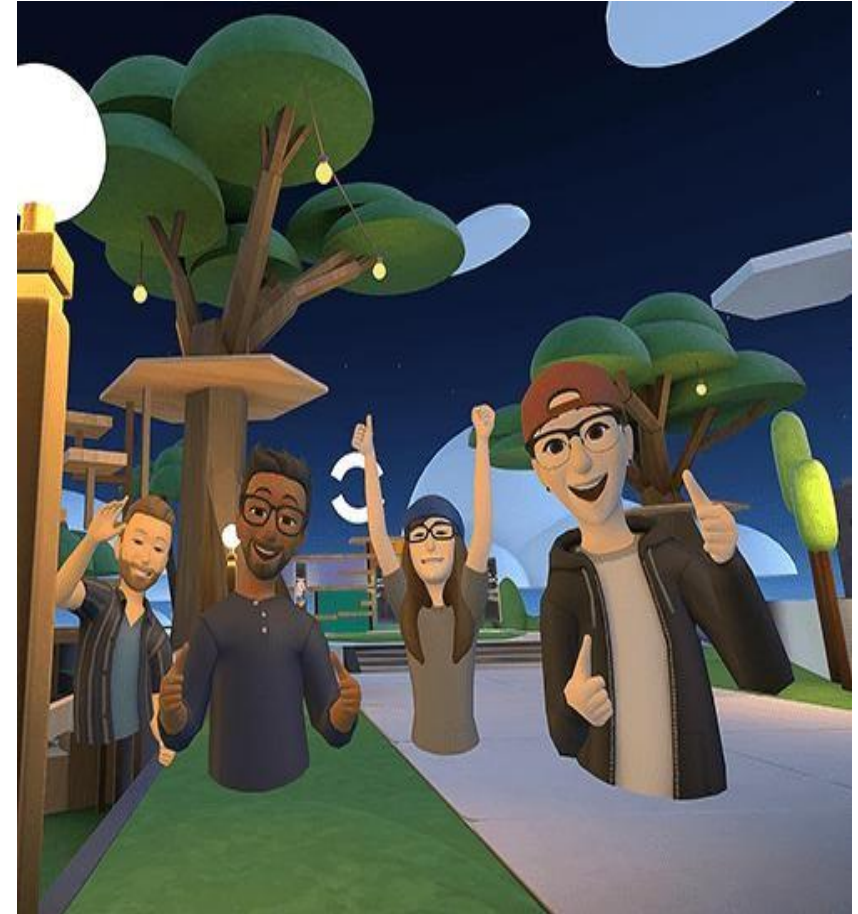
Taxonomy for Social VR Design (1)

- For social VR, researchers have spent efforts on proposing a set of **taxonomy for its application design**.
- By investigating 29 state-of-art commercial and prototypical applications, Jonas et al. first discussed the taxonomy to highlight **novel features that could enrich the social experience** ([Jonas, 2019](#)).
- The taxonomy of social VR design could fall into **three** broad aspects.
- **Self:** This aspect encapsulates features enabling users to control their avatars, including –
 - **Avatar representation** refers to how much of a physical human body is replicated in VR.
 - **Avatar customization** refers to a user's ability to modify their avatar's appearance
 - **Avatar traversal** refers to the functions to support users' movement throw their avatars (e.g., walking, flying, transporting, etc.).



Taxonomy for Social VR Design (2)

- **Interaction with others:** This aspect encapsulates features that facilitate interaction between users, including –
 - **Communication privileges** refer to ways that a user can control who is able to communicate with them, such as through user blocking and muting, friends lists, and invitations for private interaction.
 - **Communication types** refer to how users communicate, including text-based and voice communication (e.g., thought bubbles and live voice) and nonverbal communication (e.g., facial expressions, gesture, and body poses).
 - Some social VR applications **scaffold** interaction with activities. These activities can be multi-user events (e.g., public viewings of videos), recreational activities (e.g., avatar dancing), and collaborative prototyping (e.g., virtual workshops for collaboratively design and brainstorm).



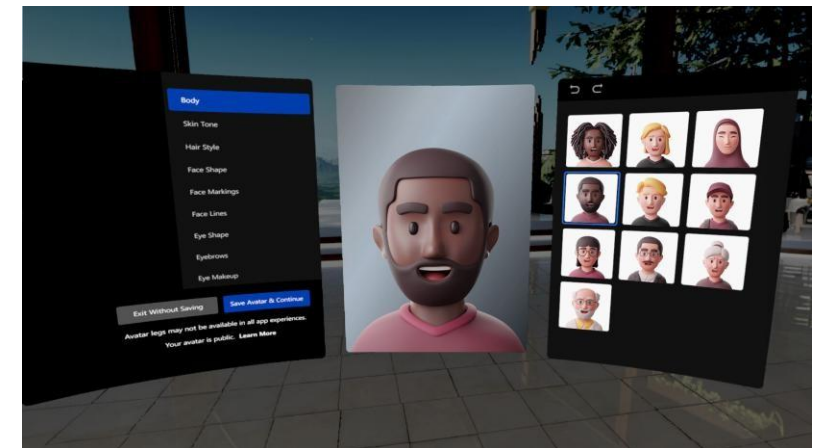
Taxonomy for Social VR Design (3)

- **The environment:** This aspect encapsulates design choices of the virtual space through which users interact and present themselves.
 - Several applications enable users to **manipulate the environment**, such as altering physical elements and creating a new virtual space.
 - Social VR applications also vary based on **spawning location**, dividing space or area for diverse usages, which could become private area (accessible only to the individual) and social area (location populated with other users).
 - In turns, the **openness of environment** refers to the capability of users to traverse different area, enabling them to discover private area (e.g., private invitation or permission) or social area (e.g., public access or community access).



Avatar and Self-presentation (1)

- As we discuss in taxonomy, **avatars** are one of the crucial components to constitute the immersion in social VR.
- In computing, an avatar is a **graphical representation** of a user or the user's character or persona.
 - Avatars can be **two-dimensional** icons in social media platforms, where they are also known as profile pictures, userpics, or formerly picons (personal icons).
 - In social VR, an avatar usually takes the form of a **three-dimensional** model that can conduct social activities like communication or interaction.
- In social VR platforms (e.g., VR Chat, Rec Room, and AltSpaceVR), avatars can be highly **expressive and interactive**. These platforms provide functions allowing users to convey a wide range of emotions and gestures.
 - Some social VR platforms even enable users to track their eye movements and facial expressions, making their avatars more lifelike and realistic.



Avatar and Self-presentation (2)

- Avatars have become an area of study in the world of academics. One major line is users' **self-presentation** through avatars.
- First, we introduce self-presentation.
- In online digital space, self-presentation refers to the way individuals **construct and present their identity or persona** through online platforms.
- On social media, self-presentation involves the **deliberate selection and display** of personal information and can take in forms of various social activities (e.g., likes, upvotes, and commenting).
- In academic communities, researchers have highlighted importance of self-presentation, as a mechanism through which online users **construct, perceive, and experience social relationship** ([Manago, 2008](#)).
- Research also has shed light on a number of practices and issues for self-presentation online, including **how to construct the authenticity and avoid duplicate of digital identities** ([DiMicco, 2007](#)).



Avatar and Self-presentation (3)

- In social VR, it is recognized that **avatars' appearance** can act as a proxy for users' self-expression, with a close relation to **users' preference and motivations** as well.
- According to an interview on social VR users (N=61) from blogging, dating, and gaming platforms, Vasalou and Joinson found that users' choices on avatars appearance could vary for their motivation in different occasions.
- In the **blogging condition**, social VR users tend to **map their personality and preferences onto their avatar**. It became apparent that the notion of “**accurate**” self-representation.
 - For instance, participants opted for their personal preference offline, such as fashion items they often wear. And hobbies, such as playing games or interests in history, were displayed on avatars as well.



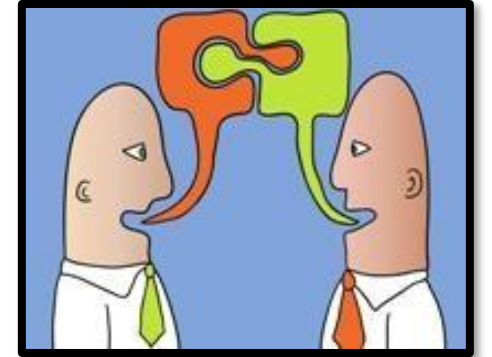
Avatar and Self-presentation (4)

- In the **dating condition**, participants were more likely to create avatars with **impressive appearance** by displaying their **attractive, or intellectual qualities**. Compared to blogging condition, users changed self-presentation strategies from “being accurate” into “*being beautified*”.
 - Even though one participant was a brunette, she changed her avatar’s hair to red as it rendered a more beautiful image for the viewer.
- In the **gaming condition**, participants developed very **diverse strategies** to self-express themselves on avatars, including being accurate or beatified as aforementioned conditions, being exaggerated to feel “cool”, or matching the story backgrounds in games.
- This study also shed light on an inspiration that, in social VR, **the nature of an online environment can prime users to create avatars that emphasize particular characteristics in self-expression**.



Self-disclosure in Social VR (1)

- While users self-express through avatars on social VR, they also perform **self-disclosure** in the meantime.
- Self-disclosure refers to revealing **unknown personal information** to others and making such information **shared knowledge** ([Joinson, 2007](#)).
- Self-disclosure can result **positive effects** in social communities.
 - For example, one of the most important benefits of self-disclosure is to **establish alliance** with others, as an essential component of **forming and maintaining social relationships** ([Stokes, 1987](#)).
 - In addition, self-disclosure can facilitate individuals to achieve self-expression, self-clarification, social validation, relationship development, and social control ([Duane, 1995](#)).



Self-disclosure in Social VR (2)

- In social VR, avatars can be viewed as the ***indicator*** towards users' self-disclosed information. Sometimes, **personal information** can be observed or inferred from self-expression, behaviors, and communications by avatars, where the disclosure could be deliberate or without attention.
- Maloney and Zamanifard first modeled users' experience and preference towards self disclosure in social VR, which can be summarized into **three** patterns ([Maloney, 2020](#)) –
 1. Social VR users often used **familiarity or anonymity** to decide to whom they chose to share information.
 2. Sharing emotions and life experiences with others was more **acceptable**.
 3. Sharing certain personal information such as gender, age, and location was **complicated**, where users' **attitudes could vary largely and become conflicting**.

Self-disclosure in Social VR (3)

- Their interview also revealed **two conflicting themes** of collective concerns regarding self-disclosure in social VR.
- Some users regarded social VR as a **risk-free** social space for freely sharing information with others, even **not** having any concerns about the existence of **privacy issues**.
 - *"I mean, besides the usual suspects like I don't say my bank information. So, I don't really have privacy issues."*
- In contrast, some others had **reservations** about sharing information and were **uncomfortable to self-disclose** in social VR. Most of these users prefer to always stay anonymous or just not sharing any kind of personal information in social VR.
 - *"I don't disclose the specifics of where I live and stuff like that."*

VRChat (1)



- Next, we talk about a mainstream social VR platform, VRChat.
- **VRChat** is a VR platform released in 2017, that allows users to create and customize avatars and interact with other users in virtual worlds.
- Nowadays, VRChat has become one of the most popular social VR platforms.
 - It is reported that there are more than **5.78 million** VRChat users in 2023 and the average number of daily active users is larger than **23,000** ([Source](#)).
- After an initial wave of viral popularity upon its release, the platform saw a steady increase in concurrent users with the onset of the **COVID-19 pandemic** and direct quarantine policies.
 - At that time, Twitch reported a record of **over 24,000** concurrent users' viewership to VRChat channels, spurred by Halloween events and the recent release of the Oculus Quest 2 ([Source](#)).

VRChat (2)



- The increasing popularity of VRChat has also led to an emerging research agenda to further investigate the **mediated interactive experiences**.
- Although VRChat was originally released as a gameplay platform, it was reported that the **social relationship** built among users could go **beyond** as game friends. **The diverse interaction on VRChat fostered different types of social connections**.
 - I.e., dating fostered imitate relationships as lovers ([Zytke, 2023](#)); collaboration in a VR workshop brought relationships as co-workers ([Saffo, 2020](#)).
- Freeman and Acena took a look on VRChat users' activities throw which users can foster close **interpersonal connections**. They found that the engagement towards shared activities in an **embodied** way could effectively form connections among users (e.g., shaking hands, hugging, etc.) ([Freeman, 2021](#)).

Harassment Issues in Social VR

- However, sometimes users' interaction on social VR can go wrong. One widely reported issue is **avatar harassment**.
- Since platforms like VRChat support **embodied behaviors**, users may receive harassment and other unwanted behavior through their experience in the virtual environment, including groping and kissing towards their avatars.
- Moreover, social VR offers a more immersive and embodied multi-user virtual environment with a **heightened sense of presence**. These novel characteristics may lead to greater instances of harassment and potentially more **destructive** consequences compared to traditional 3D virtual worlds.
- Nowadays, harassment has become a **common issue** on social VR platforms, even towards young-age users.
 - For example, posing as a 13-year-old girl on VRChat, BBC researcher Jess Sherwood said she entered a virtual strip club where she saw adult men chase a child while telling them to remove their clothes ([Source](#)).

Mitigate Social VR Harassment (1)

- Researchers also have put efforts on regulating harassing behaviors and mitigating their harmful effects to social VR communities.
- Freeman et al. conducted a survey on users (N=30) with harassment experience to **explore users' strategies and recommendations** to mitigate these new characteristics of harassment ([Freeman, 2018](#)).
- According to participants' responses, the authors summarized **three** design implications for mitigating harassment in social VR –
 - **Platform embedded voice modulators**: Modulators can identify and filter sensitive personal information in voice (e.g., gender, race, ethnicity, etc.), which can **make users' voice anonymous or generic**.
 - This helps users not disclose specific **personal cues** through voice and further avoid potential harassment based on their identity.

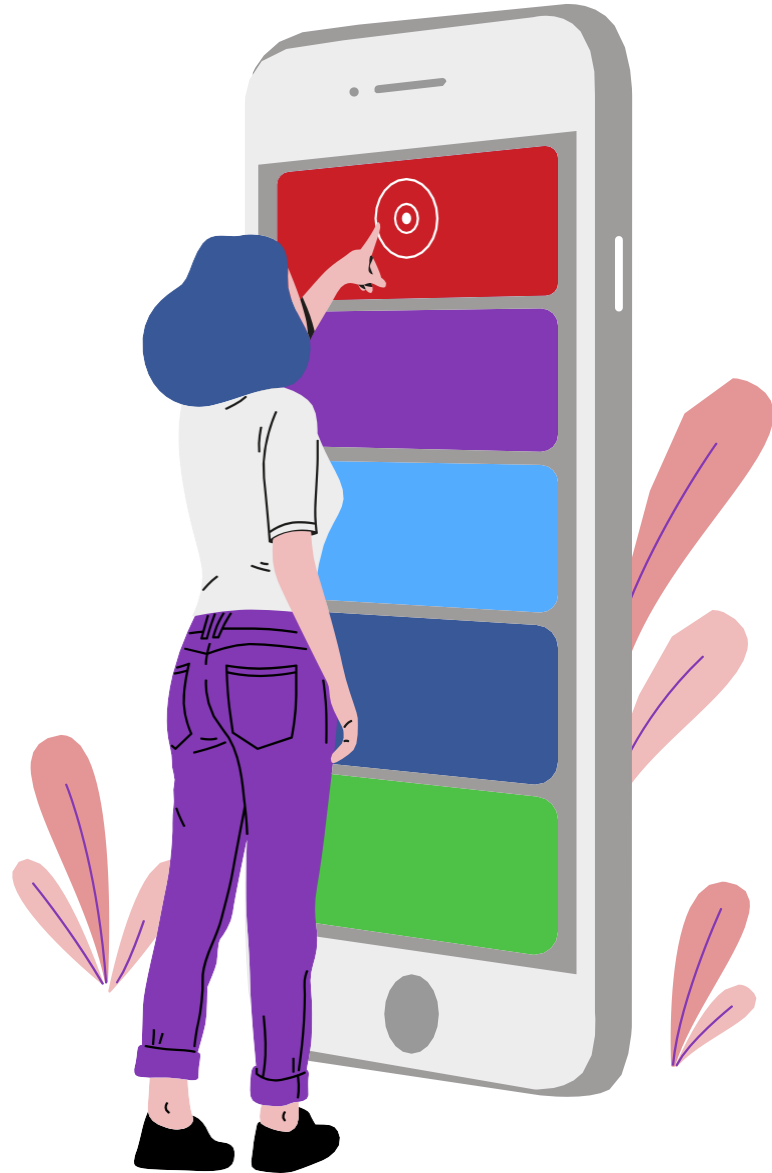
Mitigate Social VR Harassment (2)

- **More robust reporting**: Participants reported that a robust reporting system for harassment problems was necessary to mitigate the issues. Such a system majorly composed three required features –
 - **In-the-moment**: Harassment in social VR often happens within *real-time* interaction. It is important to provide users with the ability to immediately report a potential harasser **in-the-moment** (e.g., by hitting a hotkey).
 - **Detail-oriented**: Since many social VR platforms didn't have well-formed policy or regulation for harassment, it is crucial to enable users to **access and collect details** about the instance (e.g., time, location, and context), to **better trace and capture the harassers**.
 - **Necessary documentation**: Based on above two features, users also needed functions to **immediately document** an ongoing instance of harassment (e.g., hotkeys for taking screenshots or requesting witnesses).

Mitigate Social VR Harassment (3)

- **Build trust between moderators and users:** Participants also concern the **privacy** problems on communities' moderators, where their personal information could be known by a third-party.
 - Although most participants understood that the purpose of having moderators was to protect them from potential harassment, some of them seemed to be **reluctant** to be in a social VR environment that is moderated.
 - This finding highlighted the unique challenges on setting moderation in social VR communities and reflected the importance on building **a trusting mechanism** between moderators and users **in a more nuanced way**.
 - As inspired by the authors, **improving the transparency** was the most important step, fostering more organic interpersonal dynamics between moderator and users and keeping both side **in a synchronized context**.

Today's Outline



01

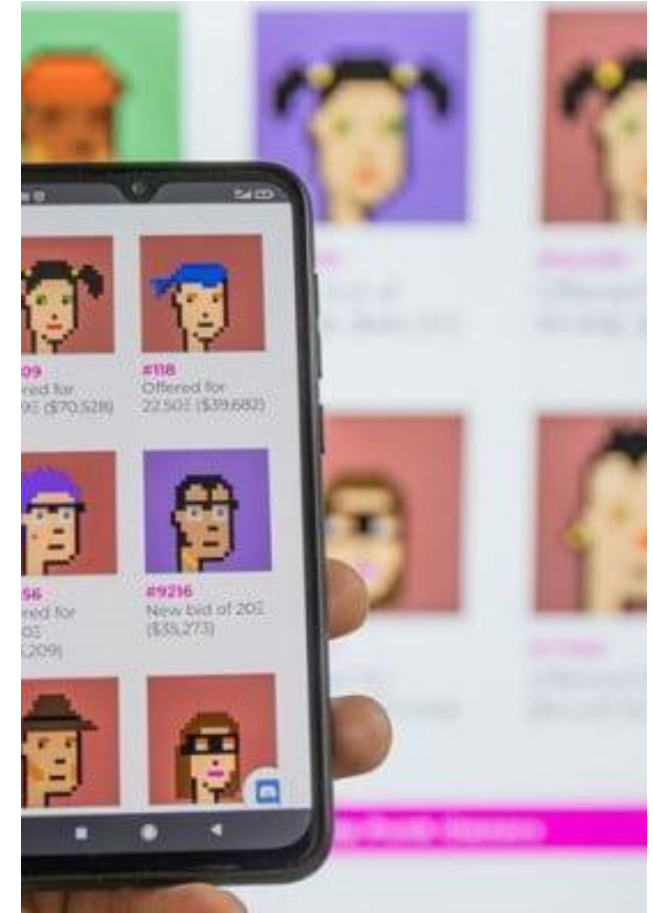
Lecturing

Extended Reality and NFT



Non-fungible token (NFT) (1)

- Finally, we discuss non-fungible token.
- A **non-fungible token (NFT)** is a unique digital identifier that **cannot be copied, substituted, or subdivided**, that is recorded in a blockchain, and that is used to certify ownership and authenticity.
- NFTs can be created by anybody and require few or no coding skills to create, which typically contain references to **digital files** such as photos, videos, and audio.
- The term “**non-fungible**” means that **the digital asset is unique and cannot be exchanged or swapped for something of equal value**.
 - In short, a thing is fungible when all equivalent amounts of that thing are interchangeable, such as cryptocurrencies.
 - I.e., \$100 fungible token can easily be exchanged for twenty \$5 bills, but NFT cannot.



Non-fungible token (NFT) (2)



- The popularity of NFT trading originated from an online game called **CryptoKitties**.
- This game was released in 2017, where players could breed, adopt and trade virtual cats as NFT products.
- Each kitty had its own DNA can lead to a **unique genetic variations**, which could not be copied by others.
- In 2017, the median price of a CryptoKitty was around **\$23.06** and an estimated **\$4.5 million** had been spent on these cartoon cats at that time ([Source](#)).
- Later, the NFT market **grew dramatically** from 2020 to 2021.
 - The most traded NFT products included music, in-game assets, art, videos, etc., and were usually bought with cryptocurrency or in dollars.
 - It was reported that the trading of NFTs in 2021 increased to more than **\$17 billion**, up by **21,000%** over 2020's total of **\$82 million** ([Source](#)).

OpenSea (1)



- **OpenSea** is one mainstream marketplace for trading NFT directly at a fixed price, or through an auction.

- Accompanied with the rapid rise of NFT marketing, OpenSea also experienced tremendous growth, becoming the largest NFT marketplace in the world by trading volume and user base.
 - In 2021, following a heightened interest in non-fungible tokens, the company's revenue reached **\$95 million** in February and **\$2.75 billion** in September of that year ([Source](#)).
 - By January 2022, the company had been valued at **\$13.3 billion** and was described as the largest non-fungible token marketplace ([Source](#)).

- OpenSea has also been at the forefront of innovation in the NFT space. The platform was one of the first to introduce **gas-free trading**, which eliminates the need for users to pay transaction fees.

OpenSea (2)



- The online communities on OpenSea also present **unique characteristics**.
- Considering **5.25 million** sales data, White et al. conducted a characterization study on OpenSea platform and revealed several interesting **patterns in NFT trading** ([White, 2022](#)).
- For **users' behaviors** in the market, the authors found that the average purchases per user had actually **decreased by 1.5 NFTs per month**. However, the overall massive transactional growth on OpenSea was **driven by a small subset of heavy-hitters**, where the users' purchases presented a **long-tail distribution**.
 - According to statistics results, the top **20%** of buyers and sellers account for **80%** of all transactions in the network, with the **top 20%** of collections making up **97.4% of the sales**.

OpenSea (3)



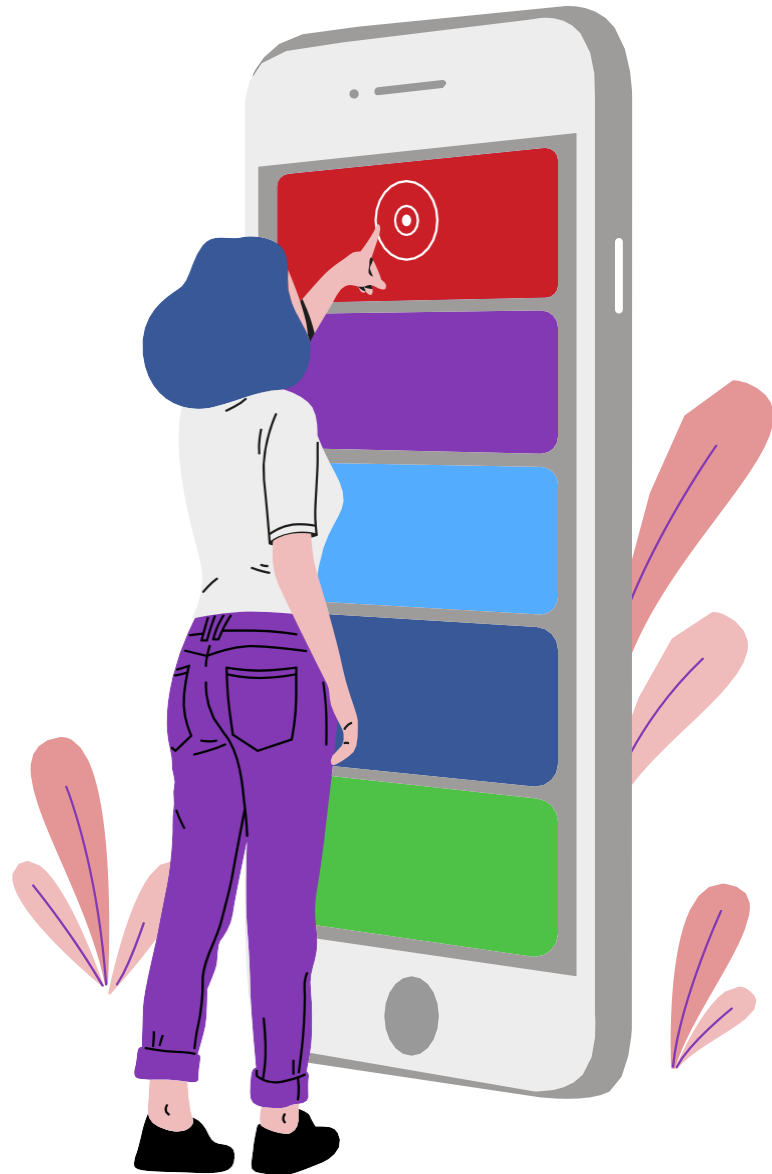
- For **popular types of NFT** purchased by users, the market orientation had shifted from a gaming focus into the art and collectible scene.
 - Of the top-100 collections sold over observed years, **88** were either **art** or **collectibles**.
 - This is accounted by that buyers sought to capitalize on the hype for **potential profit** on the **secondary sales market**.
- For **economic activities**, the authors found that the OpenSea market was **dominated by those heavy-hitters**, who contributed the most purchasing and selling amount. These top users also drove **extreme price volatility**.
 - The top 1% of buyers owned more than **108 NFTs** on average (**15x** than top 25% buyers), while the top 1% of sellers made at least **228 sales** per sellers (**22x** than top 25% sellers).
 - Those NFTs which were in the top 1% by sales value were worth upwards of **\$25,958.11** (**20x** than top 25% price), where **repeated auction** resulted in such extreme price.

OpenSea (4)



- Last but not least, the authors viewed on the trading **networks' structure** between sellers and buyers.
- Overall, the network showed obvious **sparsity**, but **communities of users were still forming** according to different categories of NFT.
 - This suggested that the level of interactivity of the average users was generally **low**, and the **cross-category buyers were not common** on OpenSea.
 - A modularity (the strength of division between communities) value of 0.282 supported the sparsity outside of these communities. Nearly 70% of user nodes had only between 1 and 5 degrees.
- Moreover, the authors also found that there were **strong interconnections between heavy-hitters** in each category's community . These powerful users tended to **congregate in these structures**.
 - This implied that the trading among heavy-hitters were **centralized**. Top buyers were more likely to purchase NFT from top sellers.

Today's Outline



03

Group Presentation

Group 5

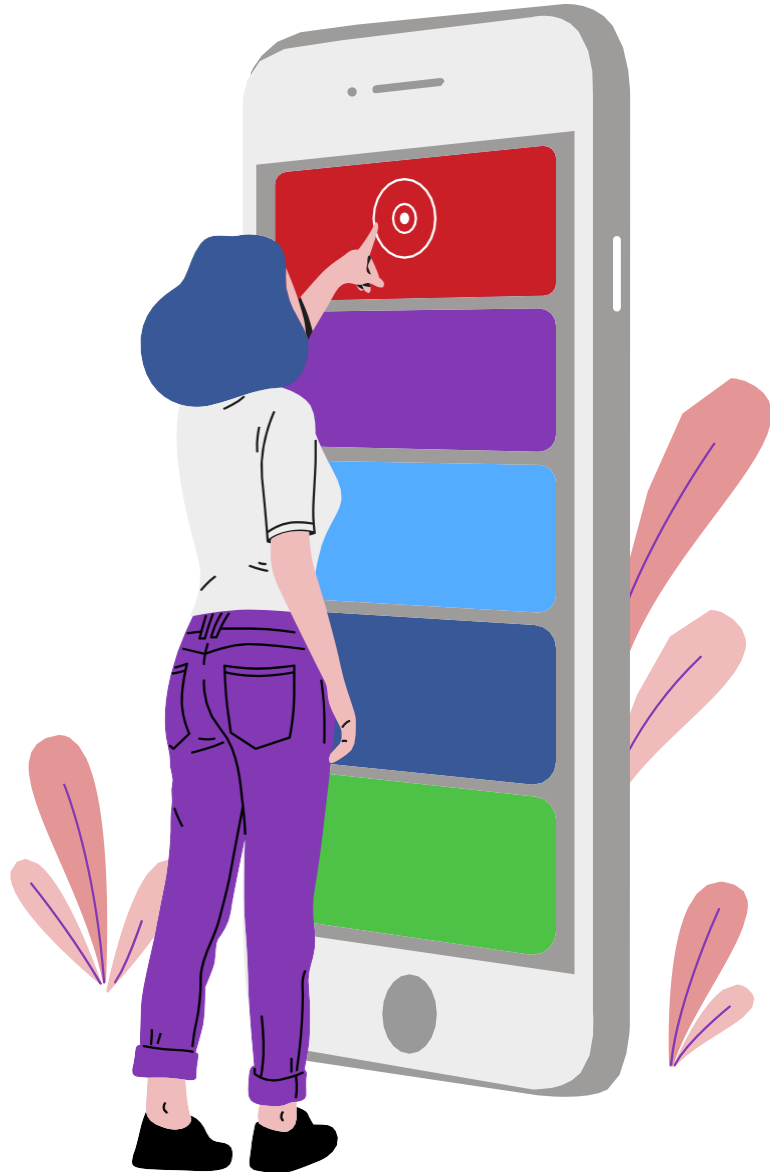


Today's Outline

03

Group Presentation

Group 5

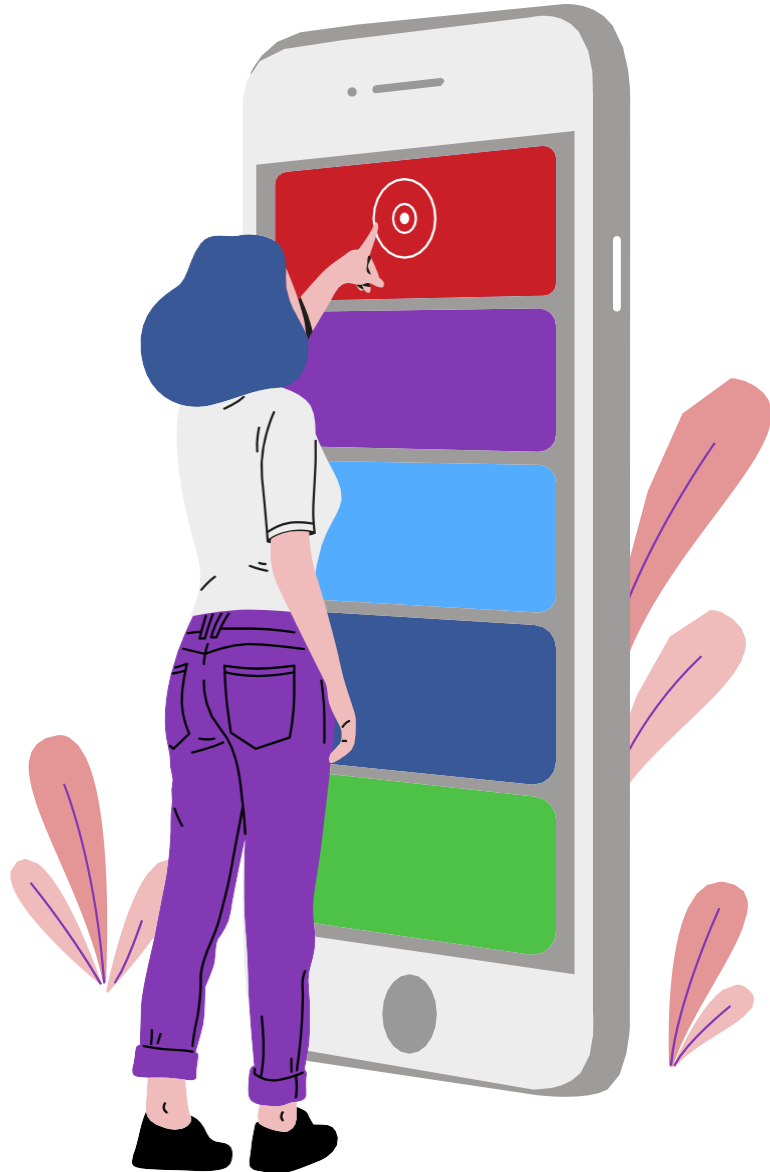


Today's Outline

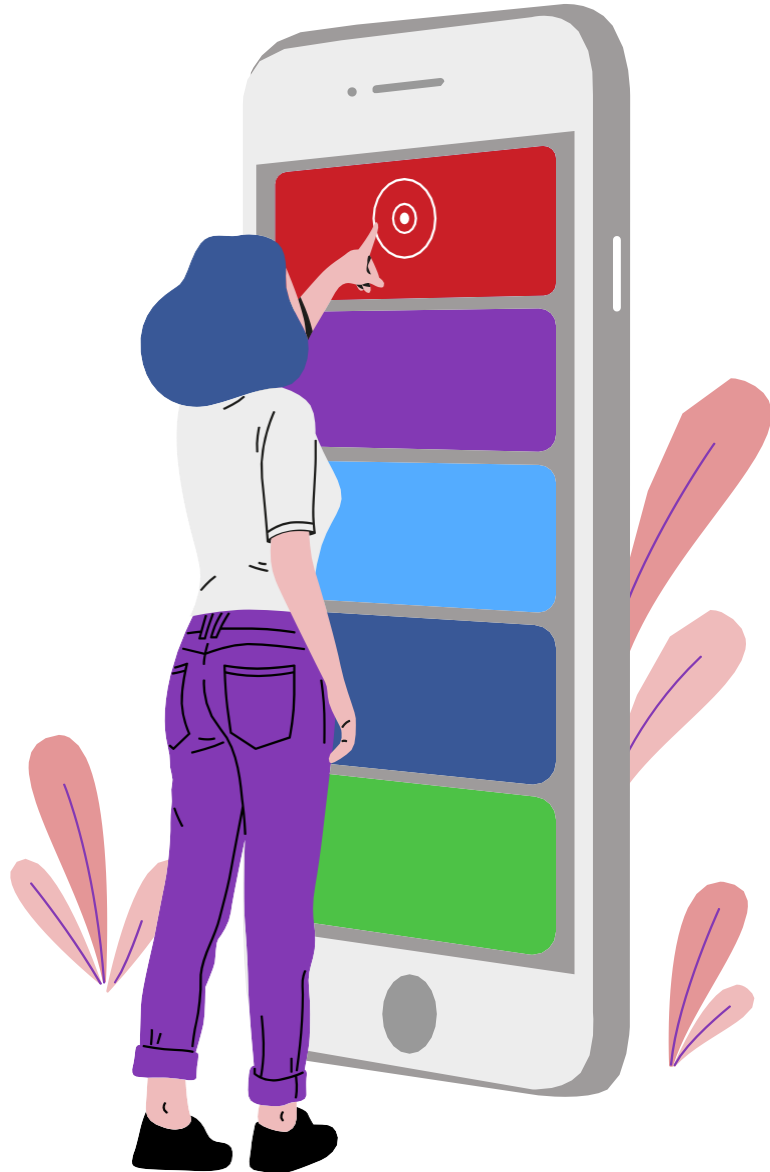
04

Group Discussion

Participation



Today's Outline



04

Group Discussion

VR Miro Time!

Task 1: Individual

Does your phenomenon have any VR/AR presence?

Member 1

Member 2

Member 3

Member 4

Are there NFTs, digital collectibles, or virtual goods?

Member 1

Member 2

Member 3

Member 4

Could XR extend this phenomenon?

Member 1

Member 2

Member 3

Member 4

One example or speculative idea.

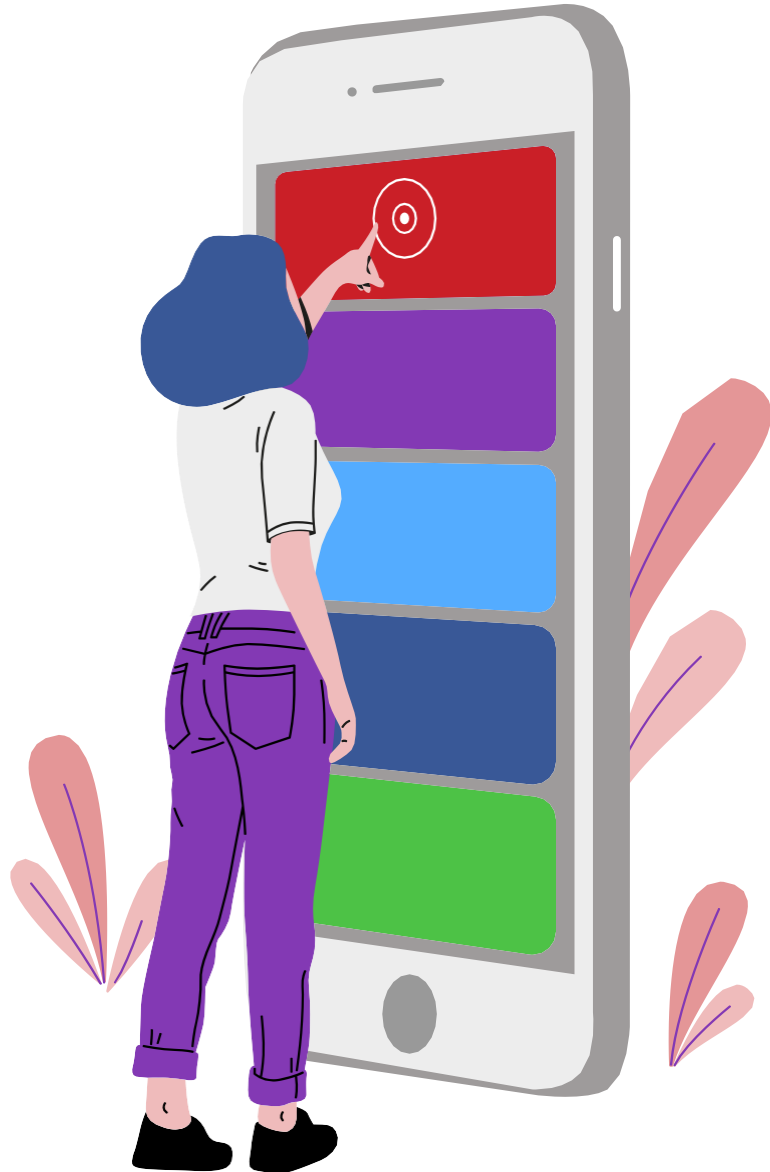
Member 1

Member 2

Member 3

Member 4

Today's Outline



04

Group Discussion

VR Miro Time!

Task 2: VR Group

How could XR transform this phenomenon?

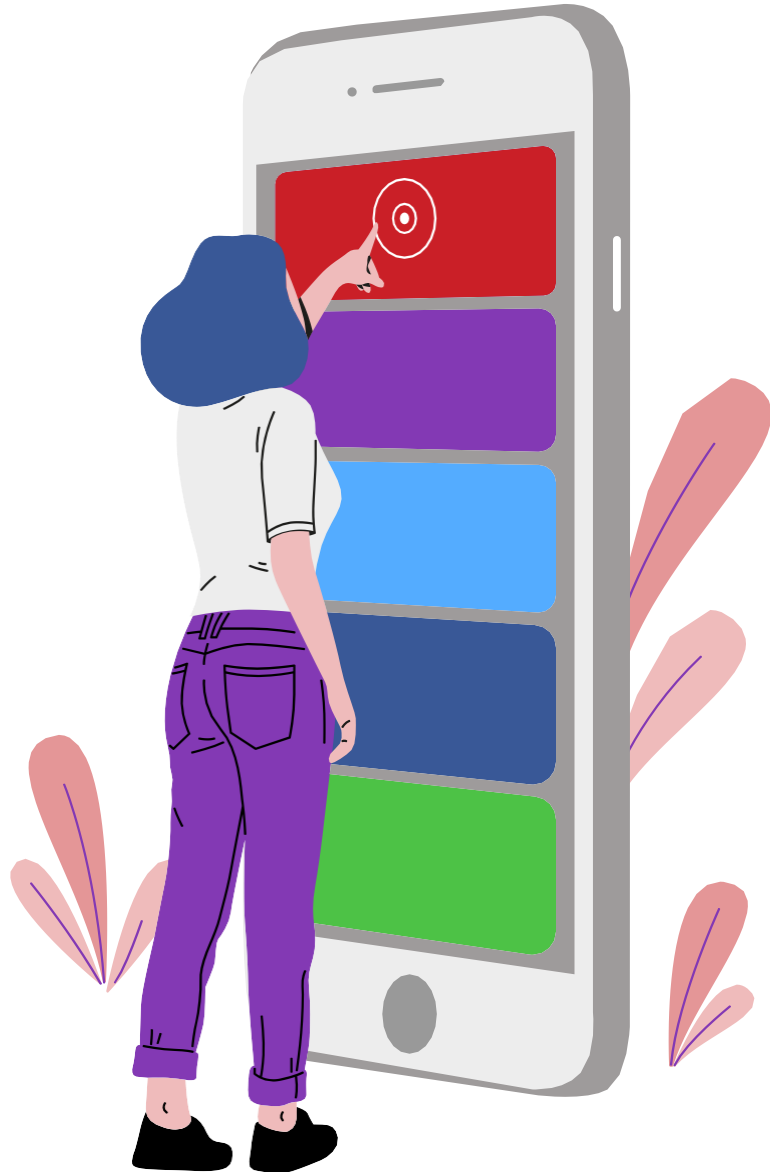
What would be gained or lost?

Are NFTs relevant here or forced?

How do attitudes toward XR/NFTs differ between HK, GZ, and Helsinki?

Identify one opportunity for XR, one skepticism or risk

Today's Outline



04

Group Discussion

VR Miro Time!

Task 3: After VR

current XR presence (if any), speculative possibilities, NFT relevance, cross-cultural attitudes toward XR, connections to previous weeks.

Answer

Answer

Answer

Answer

Answer

Answer

Answer

Answer

Individual reflection: Would XR improve this phenomenon or complicate it? What surprised you about your teammates' views?

Member 1

Member 2

Member 3

Member 4

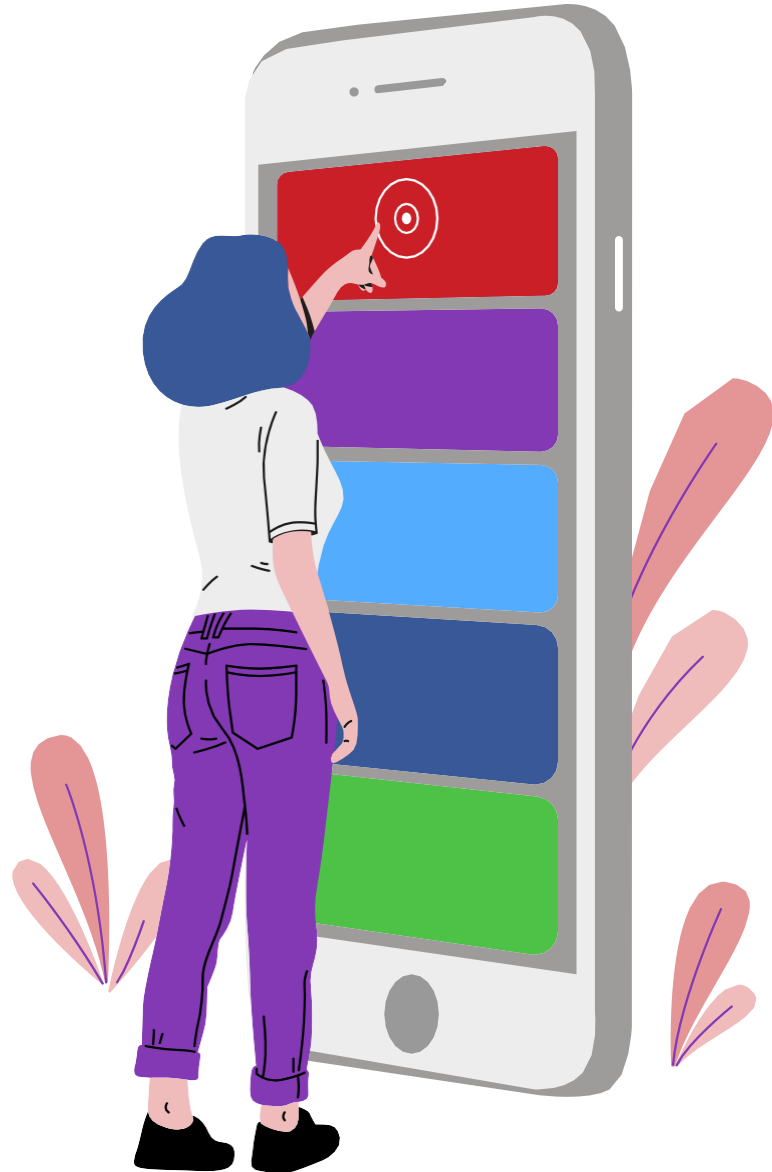
Member 1

Member 2

Member 3

Member 4

Today's Outline



05

Seminar Series

Professor Lik-Hang (Paul) Lee, PolyU

